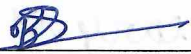


PART-B

Name of the Student: Nilamob dibrachar

Signature of the Student: ~~23~~ Nilamob dibrachar Roll No.: 230123041

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	1	7	5	2	15

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX} \\ S_{yX} & S_{XX} \end{pmatrix} \rightarrow S = \begin{pmatrix} S_{yy} & S_{yx}^T \\ S_{yx} & S_{xx} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

we know that r_1^2 is the largest eigenvalue of the matrix

$$S = S_{yy}^{-1/2} S_{yn} \cdot S_{nn}^{-1} \cdot S_{ny} \cdot S_{yy}^{-1/2} \quad S_{ny} = S_{yx}^T$$

$$4 \times 1 \times 1 \times q \times 1$$

$$S_{yy} = [S_{yy}]_{1 \times 1} \quad S_{yy}^{-1/2} = \left[\frac{1}{\sqrt{S_{yy}}} \right]$$

$$S_{yx} = [S_{y1} \dots S_{yq}]_{1 \times q} \quad S_{xx} = [S_{x1} \dots S_{xq}]_{q \times q}$$

$$\begin{bmatrix} 1 & & \\ & \dots & \\ & & q \times q \end{bmatrix}$$

$$\therefore S = \begin{bmatrix} \frac{1}{\sqrt{S_{yy}}} \end{bmatrix}_{1 \times 1} [S_{y1} \dots S_{yq}]_{1 \times q} S_{xx}^{-1} \begin{bmatrix} S_{y1} \\ \vdots \\ S_{yq} \end{bmatrix}_{q \times 1}$$

$$\Rightarrow S = \begin{bmatrix} \frac{S_{y1}}{\sqrt{S_{yy}}} & \dots & \frac{S_{yq}}{\sqrt{S_{yy}}} \end{bmatrix}_{1 \times q} S_{xx}^{-1} \begin{bmatrix} \frac{S_{y1}}{\sqrt{S_{yy}}} \\ \vdots \\ \frac{S_{yq}}{\sqrt{S_{yy}}} \end{bmatrix}_{q \times 1}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_T} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$\hat{y} = X\hat{\beta}, \quad \hat{y} = X(X^T X)^{-1} X^T y = Hy.$$

$$SS_{\text{res}} = (y - \hat{y})^T (y - \hat{y})$$

$$= \cancel{y^T y} - \cancel{y^T \hat{y}} + \hat{y}^T y (I - H)^T y (I - H)$$

$$= (I - H) y^T y (I - H) = (I - H)^2 y^T y$$

$$SS_T = \sum (y_i - \bar{y})^2 = (I - H) y^T y.$$

$\approx$

??

01

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

(a) since we need the ~~the~~ principal comp. method we selected the largest two values of eigenvalues & corresponding eigenvectors i.e.

(02)
$$\tilde{\mathbf{L}} = [\sqrt{\lambda_3} \mathbf{e}_3 \mid \sqrt{\lambda_2} \mathbf{e}_2]$$

$$\Rightarrow \tilde{\mathbf{L}} = [2.04 \mathbf{e}_3 \mid 1.44 \mathbf{e}_2]$$

(b)
$$\tilde{h}_i^2 = \sum_{j=1}^m \tilde{l}_{ij}^2 \Rightarrow \tilde{h}_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2$$

communality estimated
$$= 0.8^2 + (0.4)^2$$

$$= 0.64 + 0.16$$

(01)
$$\Rightarrow \boxed{\tilde{h}_1^2 = 0.8}$$

(c) we know that

estimated communalities
$$\tilde{h}_i^2 = \sum_{j=1}^m \tilde{l}_{ij}^2 = \tilde{l}_{i1}^2 + \tilde{l}_{i2}^2 \dots \tilde{l}_{im}^2$$

where $\tilde{l}_{ij} \rightarrow (i, j)$ th element of $\tilde{\mathbf{L}}$.

So,
$$\sum_{i=1}^p \tilde{h}_i^2 = \sum_{i=1}^p \sum_{j=1}^m \tilde{l}_{ij}^2$$

$$= \text{sum of squares of the elements of } \tilde{\mathbf{L}}.$$

$$\text{but, } \tilde{L} = [\sqrt{d_3} e_3 \mid \sqrt{d_2} e_2]$$

Notice that sum of squares of elements of $\tilde{L}$ can

$$\text{be written as } [\sqrt{d_3} e_3]^T (\sqrt{d_3} e_3) + (\sqrt{d_2} e_2)^T (\sqrt{d_2} e_2)$$

$$= d_3 \cdot e_3^T \cdot e_3 + d_2 \cdot e_2^T \cdot e_2.$$

$$\therefore \sum_{i=1}^n \tilde{h}_i^2 = d_3 \cdot e_3^T \cdot e_3 + d_2 \cdot e_2^T \cdot e_2$$

Since the given eigenvectors are normalized

$$e_3^T \cdot e_3 = 1 \text{ \& } e_2^T \cdot e_2 = 1$$

$$\therefore \sum_{i=1}^n \tilde{h}_i^2 = d_3(1) + d_2(1) = d_2 + d_3 = 6.3$$

hence, d_2 & d_3 are the largest 2 eigenvalues of S ,

hence the sum of estimated communalities can be written as largest m eigenvalues sum.

This is because in the construction of $\tilde{L}$ we took the m largest eigenvalues $\rightarrow \tilde{L} = [\sqrt{d_1} e_1 \mid \dots \mid \sqrt{d_m} e_m]$ according to principal component method.

04

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \Rightarrow E(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{n\mu}{n} = \mu$$

$$\text{var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i) = \frac{n \cdot \mu^2}{n^2} = \frac{\mu^2}{n}$$

derive $\text{var}(\bar{X}_n) = \frac{\mu^2}{n}$. { assume X_i are independent R.V.s }

all we have constant variance for the transform $y = g(\mu)$.

$$\Rightarrow [g'(\mu)]^2 \cdot \frac{\mu^2}{n} = 1 \quad [\text{for simplicity}]$$

$$\Rightarrow g'(\mu) = \frac{\sqrt{n}}{\mu}$$

$$\Rightarrow g(\mu) = \int \frac{\sqrt{n}}{u} du \Rightarrow g(\mu) = \underline{\underline{\sqrt{n} \cdot \ln \mu}}$$

05

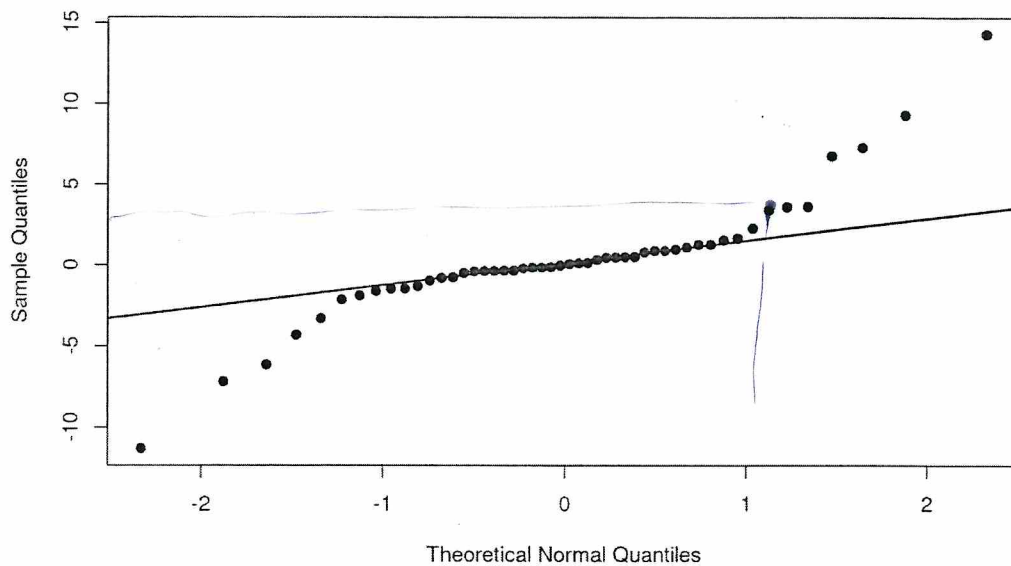
$\therefore$ the transformation $g(\bar{X}_n) = \sqrt{n} \cdot \ln \bar{X}_n$

has a constant variance of $\frac{\mu^2}{n} \cdot \left(\frac{\sqrt{n}}{\mu} \right)^2 = 1$.

$$\therefore g(\bar{X}_n) = \sqrt{n} \cdot \ln(\bar{X}_n)$$

String

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



The data does not follow a perfect normal distribution as the points do not form a straight line.

The data tends to follow a heavy tailed

distribution

data
distribution

actual Normal
distribution

02

ADDITIONAL PAGES FOR ANSWERS

